VALIDATION REPORT / 01

Are we accurate?

House of Cards: default portfolios and ten new non-tech ETF combinations

8 November 2026 | Baseline 9bdf21f | 12 portfolios, 22 ETFs, 23 distinct pairs

Holdings arithmetic passes. That is not the whole model.

The production analysis API reconciled to a separate Decimal-based calculation on all 12 portfolios. All 23 listing-level overlap calculations matched. Same-date issuer checks supported the inputs for 11 funds and 2,120 equity rows, within source rounding.

What was tested	Result	What it establishes
Portfolio/listing arithmetic	12 / 12 portfolios pass	Correct weighted aggregation on the frozen inputs
Listing overlap	23 / 23 pairs pass	Correct sum of the smaller shared weights
Independent return correlations	Largest difference 0.00031	Strong agreement with TickerInside; app has no return-correlation output
Dependency extraction replay	88.4% blind precision; 57.4% recall	Recorded evaluation, not a fresh test on the ten new portfolios

The most useful counterexample

EEM and IVV have **0.00% listing overlap**, but their independently calculated weekly return correlation is **0.675**. Shared holdings, shared suppliers and historical co-movement answer different questions.

Decision

The arithmetic is credible for the tested equity snapshots. Do not describe the app as a validated return-correlation model, assume old holdings are current, or treat missing dependency links as evidence of independence. The dependency layer needs new, blind non-tech filing evaluations before claiming general accuracy.

This is a software/data validation exercise. "Independent" means a separate calculation path and external provider/issuer controls; no second human auditor reviewed this report.

What we tested

The cohort was fixed before batch measurement; no production model or formula was tuned.

Case	Mandate / purpose	ETF allocations
D1	US core with tech tilt	IVV 40%, QQQ 25%, XLK 15%, EEM 20%
D2	Global core with Nasdaq tilt	VTI 50%, VXUS 30%, QQQ 20%
N01	Financials, large vs broad	XLF 50%, VFH 50%
N02	Healthcare, large vs broad	XLV 50%, VHT 50%
N03	Energy, large vs broad	XLE 50%, VDE 50%
N04	Utilities, large vs broad	XLU 50%, VPU 50%
N05	Consumer staples, large vs broad	XLP 50%, VDC 50%
N06	Industrials, large vs broad	XLI 50%, VIS 50%
N07	Materials, large vs broad	XLB 50%, VAW 50%
N08	Real estate, sector vs broad	XLRE 50%, VNQ 50%
N09	Financials, healthcare, utilities	XLF 40%, XLV 35%, XLU 25%
N10	Energy, staples, industrials	XLE 30%, XLP 30%, XLI 40%

D1 and D2 are the exact allocation-based defaults in hoc/examples.py. They do not use the UI's whole-share rounding. N01-N08 compare large-cap and broad funds in the same non-tech sector. N09 and N10 test mixed-sector portfolios and unequal allocations.

"New" means absent from the two default portfolios as combinations. It does not establish that an LLM has never encountered the funds or that their constituents are outside all previous development data. XLF/VFH was also used for API reconnaissance. This is a diagnostic convenience sample, not a random market sample or a training holdout.

Four separate comparison paths

Inside: pinned production sources, identity resolution, normalization, portfolio maths and /api/analysis, exercised through FastAPI TestClient. Only the HTTP input seam records/replays public source bytes; no production arithmetic is replaced.

Outside: raw holdings summed with Decimal; TickerInside public pair answers; direct iShares/SPDR/Vanguard issuer files; and Yahoo daily adjusted closes rolled into Friday-ending weeks. Pearson correlation was recomputed without application helpers and cross-checked with NumPy.

TickerInside is an upstream source for House of Cards. Agreement with its own fund files is shared-source parity, not independent truth. Direct issuer and independent price checks are reported separately.

The default ETF pairs

Company overlap and listing overlap differ by design when share classes or listings are merged.

Pair	App company overlap %	Manual listing overlap %	TI listing overlap %	Outside return r
EEM/IVV	0.0000	0.0000	0.0	0.6748
EEM/QQQ	0.2207	0.2207	0.2	0.7099
EEM/XLK	0.0000	0.0000	0.0	0.7054
IVV/QQQ	48.8247	48.8247	48.8	0.9425
IVV/XLK	35.5222	35.5222	35.5	0.8984
QQQ/VTI	48.1566	48.1566	48.2	0.9308
QQQ/VXUS	1.4203	0.5629	0.6	0.7016
QQQ/XLK	50.1063	50.1063	50.1	0.9722
VTI/VXUS	0.1362	0.1362	0.1	0.7683

Why QQQ/VXUS looks different

House of Cards reports **1.4203% company overlap**; the separate listing calculation is **0.5629%**, which TickerInside rounds to 0.6%. The 0.8574-point bridge is accounted for by ASML and Ferrovial listings:

Company	Symbols grouped by the app	Added company overlap (pp)
ASML Holding N.V.	ASML, ASML.NL	0.7896
Ferrovial SE	FER, FER.ES	0.0678

This is a definition difference, not a failed overlap formula. Company-level arithmetic was checked against the app's supplied symbol memberships; that check does not independently prove every entity match. The two listed bridges are explicitly identified for review.

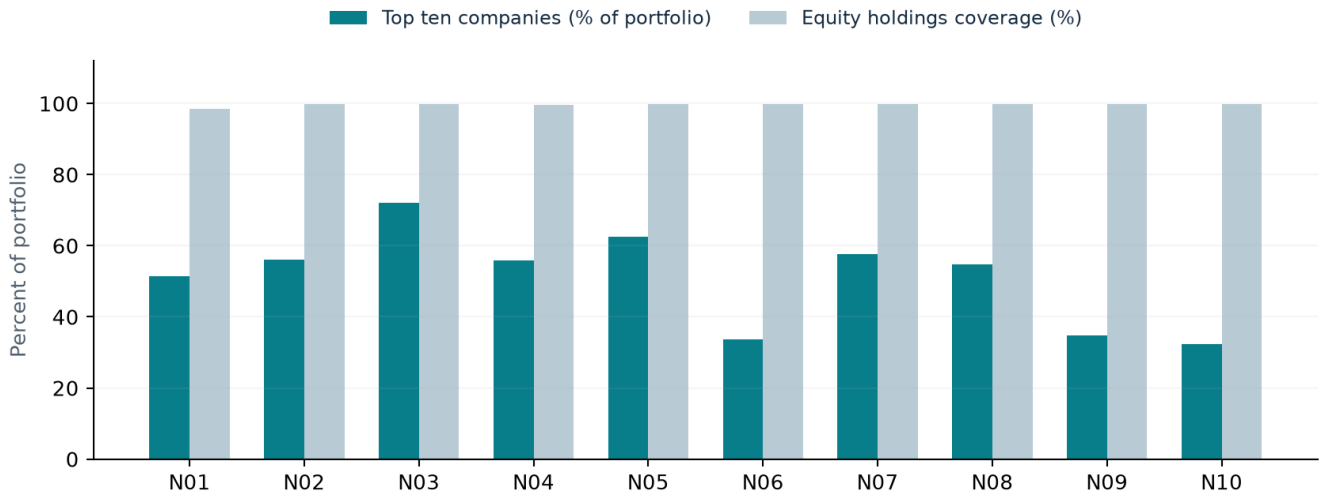
Overlap percentages compare fund holdings and do not depend on the portfolio allocation. The portfolio exposure and top-ten tests do use the allocations on page 2.

Ten new non-tech combinations

Same-sector pairs move together strongly. Mixed-sector portfolios are not proven independent by zero holdings overlap.

Case	ETF combination	App overlap %	Outside weekly r	Top ten %
N01	XLF/VFH	82.42	0.993	51.38
N02	XLV/VHT	85.35	0.991	56.00
N03	XLE/VDE	81.74	0.998	72.13
N04	XLU/VPU	88.67	0.998	55.79
N05	XLP/VDC	77.22	0.983	62.63
N06	XLI/VIS	78.35	0.991	33.74
N07	XLB/VAW	67.97	0.984	57.55
N08	XLRE/VNQ	61.49	0.992	54.76
N09	XLF/XLV/XLU	0.00	0.289-0.431	34.88
N10	XLE/XLP/XLI	0.00	0.177-0.378	32.41

For the three-fund cases N09 and N10, overlap and r show the range across their three constituent pairs, not a "portfolio correlation" or an average correlation. All correlations use the same 156 aligned weeks.



The eight same-sector tests produced high overlap and return correlations near one. The mixed-sector pairs had zero listing overlap, yet positive weekly return correlations. Concentration remains material in several same-sector portfolios, even when two ETF names are held.

Independent issuer checks

Keep current-source differences separate from calculation errors on a frozen source vintage.

Same-date control: 11 funds, 2,120 equity rows, all dated 5 November 2026. Direct iShares and SPDR weights agree with TickerInside to at most **0.000050 percentage points per line**. This is consistent with four-decimal source rounding. One immaterial IVV holding rounds out of the TickerInside map; no material holding was missing.

Pair	App / old TI %	Issuer files %	Change pp	Vanguard date in TI -> issuer
VAW/XLB	67.974	66.329	-1.645	08-31 -> 09-30
VDC/XLP	77.222	75.495	-1.727	08-31 -> 09-30
VDE/XLE	81.737	82.361	+0.624	08-31 -> 09-30
VFH/XLF	82.420	84.846	+2.426	08-31 -> 09-30
VHT/XLV	85.353	85.128	-0.225	08-31 -> 09-30
VIS/XLI	78.353	77.650	-0.703	08-31 -> 09-30
VNQ/XLRE	61.494	62.880	+1.386	07-31 -> 09-30
VPU/XLU	88.668	87.850	-0.818	08-31 -> 09-30

SPDR is 2026-11-05 on both sides. Vanguard issuer files are 2026-09-30; TickerInside uses 2026-08-31, except VNQ 2026-07-31. Direct issuer pairs use CUSIP/security identifiers. Vintage changes, corporate actions and identifier changes are therefore confounded; these deltas are not app arithmetic failures.

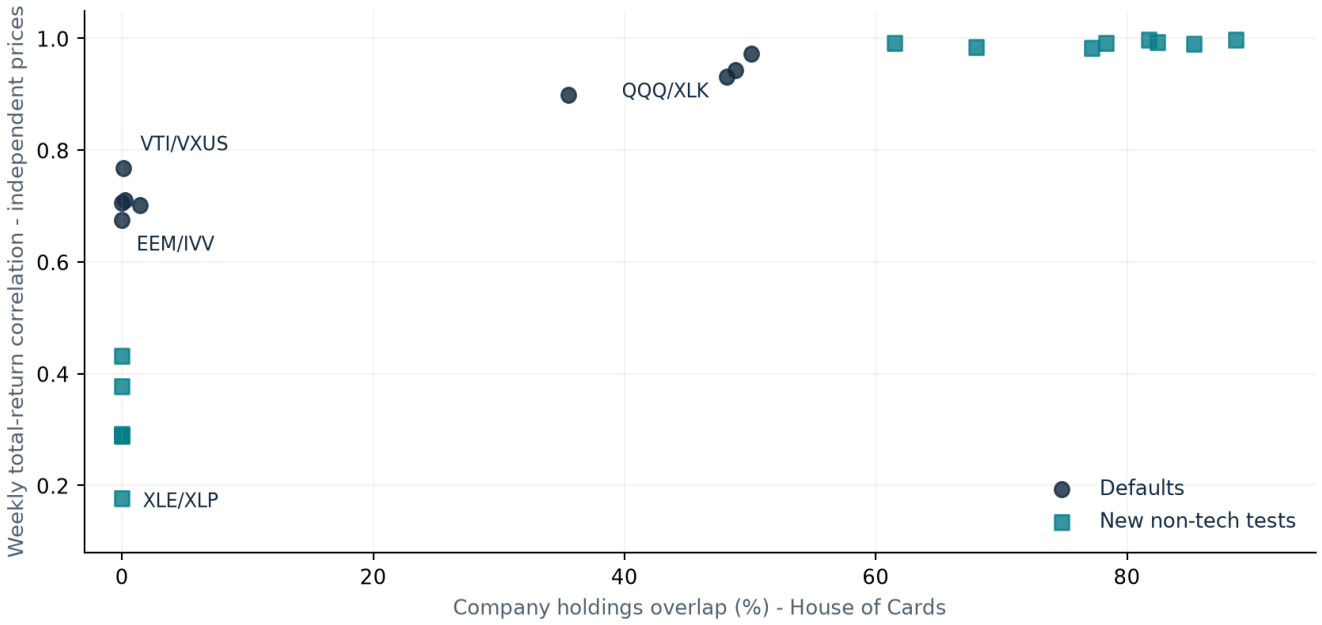
Fresh prices can sit next to old holdings

Price series end on 6 November 2026, but holdings dates range from 5 November back to 30 June. On the collection date, QQQ and VTI holdings are 131 days old, VXUS and VNQ 100 days old, and the other Vanguard sector files 69 days old. This is a real limitation of present-day exposure claims even when the calculation is exact.

Direct holdings were obtained for 21 of the 22 funds. A complete independent QQQ holdings file was not obtained, so no full issuer-control pass is claimed for it. Its independent adjusted-price correlation check is complete.

Correlation outside the app

23 pairs, 156 common Friday-ending weeks: 17 November 2023 through 6 November 2026.



Pearson correlations computed from TickerInside's published weekly total returns match its displayed pair answers within the expected two-decimal rounding bound. The largest absolute rounding difference is 0.004877. Rebuilding weekly returns from a second provider's daily dividend/split-adjusted closes changes r by at most **0.000307** across the 23 pairs.

Example	Holdings overlap %	Independent return r	Meaning
EEM/IVV	0.000	0.6748	No shared listings; substantial co-movement
VTI/VXUS	0.136	0.7683	Little overlap; common market movements
XLE/XLP	0.000	0.1766	No shared listings; lower, still positive r
QQQ/XLK	50.106	0.9722	High overlap and high co-movement

Feature gap: the production API does not calculate historical return correlation. The correlations here belong to the external benchmark; they are not predictions or outputs that our model has passed. A historical correlation layer would complement the holdings and dependency views, rather than replace them.

Weeks are aligned by date, not by truncating arrays to the same length. Both price paths are total-return adjusted. r is a historical association, not a measure of dependency direction, expected loss, or causation. The chosen pairs are not independent statistical observations.

A hand-style calculation

The independent oracle uses source weights, not the application's overlap or exposure helpers.

For N01, 50% XLF plus 50% VFH, the portfolio stake in JPMorgan is 0.5 x 11.0634% + 0.5 x 10.4679% = **10.76565%**. The app gives the same answer. Its contribution to fund overlap is the smaller fund weight, **10.4679 percentage points**; these are different calculations.

Holding	XLF %	VFH %	Manual portfolio %	App %	Overlap contribution pp
JPM	11.0634	10.4679	10.76565	10.76565	10.4679
BRK.B	13.1511	7.6840	10.41755	10.41755	7.6840
V	8.8849	4.3975	6.64120	6.64120	4.3975
MA	6.5438	5.2897	5.91675	5.91675	5.2897
BAC	3.9925	4.6092	4.30085	4.30085	3.9925

What reconciled across all twelve portfolios

Listing exposure = sum of ETF allocation x that listing's fund weight. Listing overlap = sum of min(weight A, weight B) across common listings. Top-ten concentration = the ten largest company exposures added together. HHI = sum of squared company shares after normalizing to covered company holdings; effective N = 1 / HHI.

Check	Observed maximum error	Scope
Listing exposure	0.00000000 pp	Every included listing in all 12 portfolios
Company arithmetic	0.00000000 pp	Conditional on app entity memberships
Listing overlap	0.00000000 pp	All 23 pairs
Top-ten concentration	Less than 0.00000001 pp	All 12 portfolios
HHI	5.00e-11	Within API numeric serialization precision

The equity oracle explicitly excludes nine observed index-futures contracts, including positive and negative positions. The application correctly excludes them too. Direct issuer cash, money-market rows with no equity identifier, and warrants are also separated. Raw source weights are never silently scaled to 100%.

These are independent computations by code, with a worked example suitable for manual checking. They are not a claim that a person calculated every line with a calculator. No app source or model was changed to improve the benchmark score.

How a company link is weighted

Three separate stages: discover the relationship, rank its importance, then aggregate portfolio exposure.

Detection: the backend selects portfolio companies, retrieves available SEC annual reports, and asks Gemini for named suppliers, customers, manufacturers, inputs, partners and facilities with supporting quotes. Deterministic checks ground targets and quotes in the filing; semantic interpretation review can reject or flag a relationship. Cached extraction and optional additional research sit around this pipeline.

Importance: a deterministic rule score ranks the relationship. The starting value depends on its type, then wording in the evidence changes it. This is a heuristic ranking, not a learned probability or an estimated share of production at risk.

Relationship	Starting score	Example rule
Foundry / contract manufacturer	0.90 / 0.85	High starting importance by type
Key input / supplier / joint venture	0.60 / 0.50 / 0.40	Evidence wording can raise or lower the score
Customer	0.50, 0.70 or 0.90	Grounded revenue share: under 10%, 10-20%, or at least 20%
Facility	0.70 / 0.30 / 0.50	Production / office or R&D / other location

Sole-source evidence sets a floor of 0.95. Other cues add 0.05-0.15; alternative sources subtract 0.15, with the result clipped to 0-1. A flagged interpretation or qualifying reverse-customer disclosure is capped at 0.55. Tiers are critical at 0.80+, high at 0.60+, medium at 0.40+, otherwise low.

The portfolio percentage is a different number

For a supplier, the backend sums the portfolio stakes of the distinct companies linked to that supplier. If two companies represent 2% and 3% of the portfolio, the displayed linked-company exposure is 5%, even if one filing supplies several quotes. The critical/high subset uses qualifying relationships only. The calculation does not multiply 5% by the importance score.

That 5% describes money held in linked companies. It does not establish that 5% of the portfolio would be lost, that all their revenue depends on this supplier, or that the relationship explains their return correlation. Separate supplier totals may overlap because one company can have several suppliers.

Code inspected at the frozen baseline: hoc/ai/extractor.py and verifier.py; hoc/graph/importance.py, metrics.py and orchestrator.py. The API arithmetic benchmark does not validate the economic calibration of these relationship scores. The recorded importance evaluation follows on the next page.

What the existing AI eval says

Replay of the pinned ten-company extraction benchmark, not a new inference run on the ETF portfolios.

Metric	Blind gold	With pooled additions
Precision (95% company bootstrap CI)	88.4% (83.2-92.5)	96.1% (89.6-100.0)
Core recall (95% CI)	57.4% (36.8-76.6)	57.4% (35.3-76.0)
F1 (95% CI)	69.6% (51.9-82.7)	71.9% (53.3-85.1)
Predictions / core misses	129 / 49	129 / 49

The blind gold contains 212 relationships. Ten additional relationships were accepted through pooling after reviewing predictions, making 222. Pooled precision is useful, but the blind 88.4% estimate is the stronger headline for generalization. Core recall is 66/115, so 49 core relationships are missed.

Additional measure	Result	Interpretation
Importance exact agreement	68.4% (54/79)	Tier labels disagree on roughly a third of scored matches
Importance weighted kappa	0.475 (95% CI 0.382-0.631)	Useful signal, not calibrated economic dependence
Direction / facility-country agreement	124/124 and 39/39	Conditional on evaluated matches; not whole-graph accuracy
Agentic research effect	35 findings; 0 edges added or removed	No measured precision or recall improvement on this replay

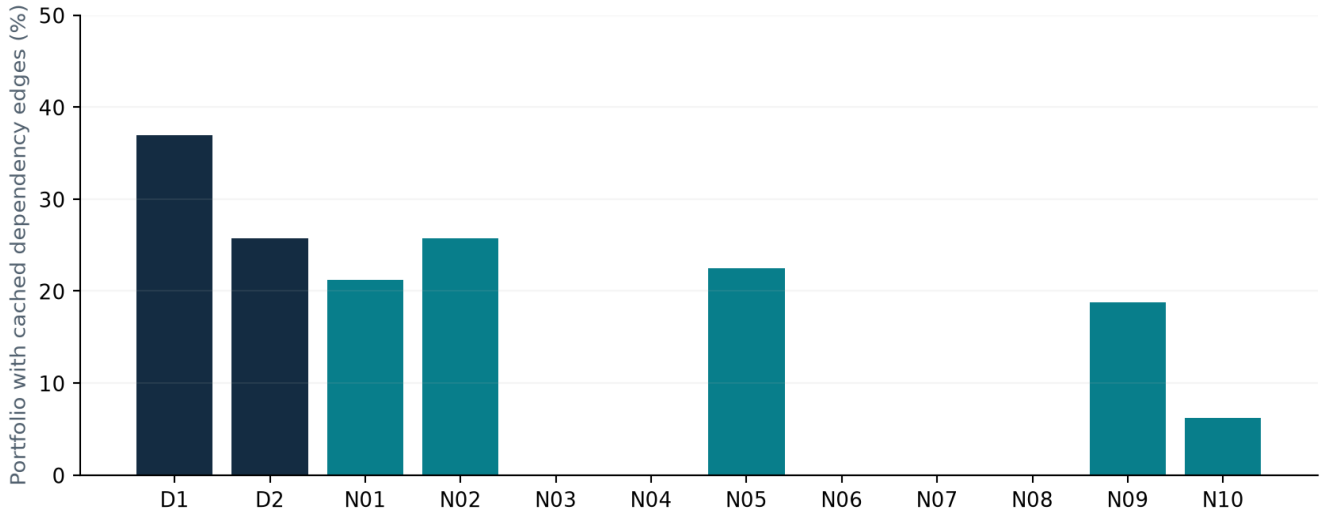
Failures matter more than a single headline

Company	Core found / missed	Blind precision	Core recall
PFE	1 / 18	100.0%	5.3%
GM	3 / 6	85.7%	33.3%
BA	5 / 7	80.0%	41.7%
NKE	5 / 4	66.7%	55.6%

Source: existing gold and recorded predictions in the repository; replay run at baseline 9bdf21f, B=2,000 company bootstrap draws. This result evaluates stored predictions, not whether newer extractor/scoring code would produce the same results in a fresh live run. Extended gold is optional for recall; these are not exhaustive filing labels.

What remains unvalidated

Cache availability is a coverage check, not a test of what the AI would discover in a fresh scan.



Case	Top companies	Cached companies	Portfolio with cached links %
D1	15	14	36.96
D2	15	15	25.71
N01	15	2	21.18
N02	15	3	25.75
N03	15	0	0.00
N04	15	0	0.00
N05	15	2	22.49
N06	15	0	0.00
N07	15	0	0.00
N08	15	0	0.00
N09	15	4	18.81
N10	15	2	6.23

The local extraction-cache snapshot contains no top-company results for five of the ten new non-tech portfolios. That does not mean those companies have no dependencies, and it does not predict coverage after a fresh scan. It shows why the holdings benchmark alone cannot validate the dependency graph on these combinations.

No fresh LLM extraction or new blind filing annotation was run for this cohort. The requested overlap/correlation benchmark is complete; non-tech semantic extraction accuracy remains an explicitly separate, unmeasured question. Cache numbers are time-specific local state, with the selected cached edge payloads saved for inspection.

What to change next

Priorities follow the evidence, rather than tuning the model to these twelve portfolios.

Priority	Action	Acceptance criterion
1	Name the metric at every comparison	Company overlap, listing overlap, return correlation and dependency exposure are separate labels, with their own dates and units.
1	Make stale holdings conspicuous	Show holdings vintage separately from the latest price; prefer available fresh issuer data or explain the older source selection.
1	Expand blind non-tech dependency evaluation	Freeze new filings/labels before running inference; score missed links, incorrect links, direction, importance and portfolio exposure coverage.
2	Add correlation only as a distinct feature	Adjusted returns, weekly date alignment, minimum sample size and source/window shown; validate against the external series retained here.
2	Retain source and calculation regression checks	Run the frozen cases without modifying them; treat version/date changes as new benchmark runs.

Keep the correct parts

Do not modify holdings arithmetic to force company overlap to equal TickerInside's listing overlap. Do not multiply portfolio exposure by the heuristic importance score and call it measured loss. The app's source aggregation and calculations passed the checks designed for them.

Verification and limits

The focused backend test run passed 152 tests with one skipped live test. Existing evaluation replay completed. The initial test command needed a task-local temporary directory because Windows denied access to the shared pytest temp directory; the rerun passed without changing tests.

The API run used captured TickerInside inputs and the real normalization/identity/maths implementation. It did not test upstream outage retries, user-interface rendering, intraday quotes, trading outcomes, live dependency extraction on the new portfolios, or completeness of every company alias. The sample is deliberately diagnostic and cannot establish universal accuracy.

The two independent price providers may use shared exchange/vendor feeds. Their agreement is an independent provider check, not proof of wholly independent market observations. Issuer files were kept locally for verification and are not redistributed in the evidence bundle.

All 23 pair results

Detailed numerical results; full precision and dates are in the companion CSV.

Pair	App company %	Manual listing %	TI listing %	TI r (shown)	Manual TI r	Outside r
EEM/IVV	0.0000	0.0000	0.0	0.67	0.67488	0.67484
EEM/QQQ	0.2207	0.2207	0.2	0.71	0.70983	0.70994
EEM/XLK	0.0000	0.0000	0.0	0.71	0.70541	0.70539
IVV/QQQ	48.8247	48.8247	48.8	0.94	0.94245	0.94247
IVV/XLK	35.5222	35.5222	35.5	0.90	0.89836	0.89840
QQQ/VTI	48.1566	48.1566	48.2	0.93	0.93081	0.93083
QQQ/VXUS	1.4203	0.5629	0.6	0.70	0.70156	0.70159
QQQ/XLK	50.1063	50.1063	50.1	0.97	0.97218	0.97222
VAW/XLB	67.9741	67.9741	68.0	0.98	0.98424	0.98424
VDC/XLP	77.2218	77.2218	77.2	0.98	0.98334	0.98335
VDE/XLE	81.7365	81.7365	81.7	1.00	0.99797	0.99797
VFH/XLF	82.4203	82.4203	82.4	0.99	0.99344	0.99344
VHT/XLV	85.3531	85.3531	85.4	0.99	0.99095	0.99097
VIS/XLI	78.3531	78.3531	78.4	0.99	0.99141	0.99143
VNQ/XLRE	61.4938	61.4938	61.5	0.99	0.99146	0.99177
VPU/XLU	88.6678	88.6678	88.7	1.00	0.99756	0.99758
VTI/VXUS	0.1362	0.1362	0.1	0.77	0.76838	0.76832
XLE/XLI	0.0000	0.0000	0.0	0.29	0.28891	0.28880
XLE/XLP	0.0000	0.0000	0.0	0.18	0.17677	0.17656
XLF/XLU	0.0000	0.0000	0.0	0.29	0.29159	0.29180
XLF/XLV	0.0000	0.0000	0.0	0.43	0.43140	0.43137
XLI/XLP	0.0000	0.0000	0.0	0.38	0.37805	0.37809
XLU/XLV	0.0000	0.0000	0.0	0.29	0.28876	0.28885

All independent correlations use 156 observations. The production app has no return-correlation result to compare. The TickerInside overlap endpoint refused large global pairs with HTTP 400 "too_large"; its documented diversify endpoint supplied the equivalent pair figures. Both the refusal and alternate response were retained.

Sources and reproducibility

The report is backed by source snapshots, hashes, independent calculations and machine-readable results.

TickerInside API and schemas

TickerInside overlap definition

TickerInside calculation methodology

Example raw fund: XLF

Example pair: XLF / VFH

Official SPDR holdings: XLF

Official Vanguard holdings: VFH

Official iShares holdings: IVV

Official iShares holdings: EEM

Independent adjusted-price history: XLF

TickerInside data is credited to TickerInside, holdings/returns through 2026-11-06, CC BY 4.0. Every fund/pair endpoint and retrieval timestamp is in sources.csv; the local snapshot manifest includes content SHA-256 hashes. Issuer source dates are carried individually, not inferred from the retrieval date.

Code and artifacts

Source: backend/tests/live/etf_accuracy/. The scripts are run_benchmark.py, analyze_results.py and build_pdf.py; cases.json fixes the cohort. The frozen baseline commit is 9bdf21f88fff6f2b260cf76e1547a63915c34975.

Reproduce with the baseline backend installed and report dependencies available: run run_benchmark.py all; then analyze_results.py; then build_pdf.py. Network responses are cached under artifacts/raw, once per URL. Preserve these bytes to reproduce this run; fetching newer upstream data creates a new data vintage.

Delivered companion file	Contents
ETF-benchmark-evidence.zip	Case manifest, portfolio/pair results, dates, issuer controls, cache inventory, source URL/hash manifest, validation summary and scripts
ETF-pair-comparisons.csv	All 23 pairs, full precision and source dates
ETF-portfolio-checks.csv	All 12 portfolio arithmetic checks

Locally retained evidence includes full API responses, source bytes, replay evaluation output, selected extraction-cache payloads and verification logs. The compact delivery bundle omits raw issuer holdings and secrets. No API keys or account information were used in the report.